



Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS



Academic Year 2026-27
FY MCA- SEM I
IT12 Data Structure and Algorithms
Practice Assignment1

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CODE:-

```
class Node:
    def __init__(self,data=None):
        self.data=data
        self.next=None
class SLL:
    def __init__(self):
        self.size=0
        self.head=None

    def append(self,data):
        newnode=Node(data)
        self.size+=1
        if not self.head:
            self.head=newnode
            return
        temp=self.head
        while temp.next:
            temp=temp.next
        temp.next=newnode

    def traverse(self):
        if not self.head:
            print("LL is empty")
            return
        print("List data:")
        temp=self.head
        while temp:
            print(temp.data,end=" ")
            temp=temp.next

    def insertAtPos(self,data,pos):
```



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```
        if pos<1 or pos>self.size or self.head==None:
            print("insertion not possible")
            return
        newnode=Node(data)
        tc=self.head
        tpre=tc
        c=0
        while c!=pos and tc.next!=None:
            tpre=tc
            tc=tc.next
            c+=1
        tpre.next=newnode
        newnode.next=tc
        self.size+=1

def middleNode(self):
    mid = self.size // 2
    temp=self.head
    while mid!=0:
        temp = temp.next
        mid -= 1
    print("\nMid Node Val:",temp.data)

def deleteAtPos(self,pos):
    if pos<1 or pos>self.size or self.head==None:
        print("list empty")
        return
    if pos==1:
        self.head=self.head.next
        self.size-=1
        return
    tc=self.head
    tpre=tc
    c=1
    while c<pos:
        tpre=tc
        tc=tc.next
        c+=1
    tpre.next=tc.next
    self.size-=1
```



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```
def reverseList(self):
    pre=None
    crnt=self.head
    while crnt:
        nextnode=crnt.next
        crnt.next=pre
        pre=crnt
        crnt=nextnode
    self.head=pre
    print("List reversed")
```

```
def sum2Consecutive(self):
    if self.head==None or self.head.next==None:
        print("\nInsufficient no. of nodes")
        return
    print("\nSum of every 2 consecutive nodes:")
    temp = self.head
    while temp.next:
        print(temp.data + temp.next.data, end=" ")
        temp = temp.next
```

```
ml=SLL()
```

```
for i in range(1,7):
    ml.append(i)
```

```
ml.traverse()
ml.middleNode()
print("size:",ml.size)
ml.reverseList()
ml.traverse()
ml.sum2Consecutive()
```

SCREENSHOT:

```
List data:
1 2 3 4 5 6
Mid Node Val: 4
size: 6
List reversed
List data:
6 5 4 3 2 1
Sum of every 2 consecutive nodes:
11 9 7 5 3
```



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